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NFS

Task 1

Server Side

Let's install nfs-utils package in server side:

```
[root@server ~ #] dnf install nfs-utils -y
Last metadata expiration check: 1 day, 1:36:41 ago on Tue 1
Dependencies resolved.
=====
Package                                Architecture
```

Let's add these services to firewall permanently:

```
[root@server ~ #] firewall-cmd --permanent --add-service=nfs
success
[root@server ~ #] firewall-cmd --permanent --add-service=rpc-bind
success
[root@server ~ #] firewall-cmd --permanent --add-service=mountd
success
[root@server ~ #] firewall-cmd --reload
success
```

Let's create a shared directory:

```
[root@server ~ #] mkdir /nfsshare
[root@server ~ #] echo "Testing Purpose" > /nfsshare/example.txt
```

```
[root@server ~ #] chmod o+rwxt /nfsshare #Not mandatory
[root@server ~ #]
```

Then we should update /etc/exports file:

```
[root@server ~ #] vim /etc/exports
[root@server ~ #] cat /etc/exports
/nfsshare *(rw)
```

Let's enable and start the service:

```
[root@server ~ #] systemctl enable --now nfs-server.service
Created symlink /etc/systemd/system/multi-user.target.wants/nfs-server.service → /usr/lib/systemd/system/nfs-server.service.
[root@server ~ #]
```

Client Side

Initially we should download the package here, too:

```
[root@client ~]# dnf install nfs-utils -y
Updating Subscription Management repositories:
```

Now we can check whether any directory is shared or not by entering server's IP:

```
[root@client ~]# showmount -e 192.168.88.160
Export list for 192.168.88.160:
/nfsshare *
```

Let's create a mountpoint:

```
192.168.88.160:/nfsshare /test
[root@client ~]# mkdir /test
[root@client ~]# tail -n1 /etc
```

We need to open /etc/fstab file and add a line here:

```
[root@client ~]# tail -n1 /etc/fstab
192.168.88.160:/nfsshare /test nfs defaults,_netdev 0 0
[root@client ~]# systemctl daemon-reload && mount -a
[root@client ~]#
```

Now we enable and start the service:

```
[root@client ~]# systemctl enable --now nfs-server
Created symlink '/etc/systemd/system/multi-user.target.wants/nfs-server.service' → '/usr/lib/systemd/system/nfs-server.service'.
[root@client ~]#
```

When we check our /test directory in client side:

```
[root@client ~]# ll /test
total 4
-rw-r--r--. 1 root root 16 May 13 01:59 example.txt
```

The file is from server.

Task 2

Server Side

First let's design server side:

```
[root@server ~ #] dnf install -y nfs-utils autofs
Last metadata expiration check: 1:11:01 ago on Tue 12 May 2026 12:19:15 AM +04.
Dependencies resolved.
```

```
[root@server ~ #] firewall-cmd --permanent --add-service=nfs
success
[root@server ~ #] firewall-cmd --permanent --add-service=rpc-bind
success
[root@server ~ #] firewall-cmd --permanent --add-service=mountd
success
[root@server ~ #] firewall-cmd --reload
success
[root@server ~ #] firewall-cmd --list-services
cockpit dhcpv6-client mountd mysql nfs ntp rpc-bind samba ssh
```

Let's create a directory and add a line to /etc/exports:

```
[root@server ~ #] mkdir /users
[root@server ~ #] vim /etc/exports
[root@server ~ #] cat /etc/exports
/users *(rw)
```

Then enable and start the services:

```
[root@server ~ #] systemctl enable --now nfs-server.service
Created symlink /etc/systemd/system/multi-user.target.wants/nfs-server.service → /usr/lib/systemd/system/nfs-server.service.
[root@server ~ #] systemctl enable --now autofs.service
Created symlink /etc/systemd/system/multi-user.target.wants/autofs.service → /usr/lib/systemd/system/autofs.service.
```

Client Side

Let's download the packages:

```
[root@client ~]# dnf install -y nfs-utils autofs
```

Then we open this file:

```
[root@client ~]# vim /etc/auto.master
[root@client ~]#
```

And add this:

```
/etc /etc/auto.master
/users /etc/auto.users
#
```

Then create `/etc/auto.users` file and add this:

```
[root@client ~]# cat /etc/auto.users
ravan -rw, sync 192.168.88.160:/users
```

The ip is the ip of server.

When we check:

```
[root@client ~]# showmount -e 192.168.88.160
Export list for 192.168.88.160:
/users *
```

Now let's enable and start the services:

```
[root@client ~]# systemctl enable --now nfs-server.service
Created symlink '/etc/systemd/system/multi-user.target.wants/nfs-server.service' → '/usr/lib/systemd/system/nfs-server.service'.
[root@client ~]# systemctl enable --now autofs
Created symlink '/etc/systemd/system/multi-user.target.wants/autofs.service' → '/usr/lib/systemd/system/autofs.service'.
```

Now let's check:

```
[root@client ~]# cd /users
[root@client users]# ll
total 0
[root@client users]# cd ravan
[root@client ravan]# ll
total 0
```

As you can see, initially there is no directory named `ravan` inside `/users`. When I run `cd ravan`, it is auto created and even if switched to this directory.

NTP

Task 1

Server Side

Timezone:

```
[root@server ~ #] timedatectl set-timezone Asia/Baku
```

Let's download the package and enable and start its service:

```
ERROR: Unable to find a match for chrony
[root@server ~ #] dnf install chrony -y
Last metadata expiration check: 0:21:37
```

```
[root@server ~ #] systemctl enable --now chronyd
Created symlink /etc/systemd/system/multi-user.target.wants/chronyd.service → /usr/lib/systemd/system/chronyd.service.
```

We should add ntp to firewall:

```
[root@server ~ #] firewall-cmd --add-service=ntp
success
[root@server ~ #] firewall-cmd --add-service=ntp --permanent
success
[root@server ~ #] firewall-cmd --reload
success
[root@server ~ #] firewall-cmd --list-services
cockpit dhcpv6-client mountd nfs ntp rpc-bind samba ssh
```

I keep pool the same in /etc/chrony.d. I just need to uncomment this:

```
# Allow NTP client access from local network.
allow 192.168.0.0/24
```

Let's set ntp true and restart service:

```
[root@server ~ #] timedatectl set-ntp true
[root@server ~ #] systemctl restart chronyd
```

Client Side

We should install the same package in Client Side, too:

```
[root@client ~]# dnf install -y chrony
```

Service enable and start:

```
[root@client ~]# systemctl enable --now chronyd
Created symlink '/etc/systemd/system/multi-user.target.wants/chronyd.service' → '/usr/lib/systemd/system/chronyd.service'.
```

We need to add ntp to firewall:

```
[root@client ~]# firewall-cmd --add-service=ntp
success
[root@client ~]# firewall-cmd --add-service=ntp --permanent
success
[root@client ~]# firewall-cmd --reload
success
[root@client ~]# firewall-cmd --list-services
cockpit dhcpv6-client ntp ssh
```

In `/etc/chrony.conf` file we need to write this line (ip is server's ip):

```
# Please consider joining the
server 192.168.88.160 iburst
# Use NTP servers from DHCP
```

We set true ntp and restart the service:

```
[root@client ~]# timedatectl set-ntp true
[root@client ~]# systemctl restart chronyd
```

Let's check it in client:

```
[root@client ~]# chronyc sources
MS Name/IP address         Stratum Poll Reach LastRx Last sample
=====
^* 192.168.88.160           3      6   17   38  +6514ns[ +62us] +/-  30ms
[root@client ~]#
```

Because there is `\*`, it means everything works well.